

- Description

This module manages an industrial pressure monitoring system, ensuring that:

1. **The pressure level in a pipeline or tank is continuously measured and compared against safety limits.**
2. **If the pressure exceeds the maximum threshold, a high-pressure warning is triggered.**
3. **If the pressure drops below the minimum threshold, a potential leakage alert is generated.**
4. **A log of all pressure readings, warnings, and corrective actions is maintained for safety compliance.**

This system helps prevent pressure-related hazards and ensures the integrity of industrial pipelines and tanks.

- Specification

module IndustrialPressureMonitoring

var currentPressure, maxPressure, minPressure: real

var pressureLog: sequence of string

process updatePressure(changeAmount: real)

functional scenario form:

currentPressure $\geq$ minPressure && pressureLog = pressureLog $\oplus$ ("Pressure Updated: " + currentPressure + " bar"), Pressure updated successfully.

||

currentPressure > maxPressure && pressureLog = pressureLog $\oplus$ ("High Pressure Warning: " + currentPressure + " bar"), High pressure warning triggered.

||

currentPressure < minPressure && pressureLog = pressureLog $\oplus$ ("Leakage Alert: " + currentPressure + " bar"), Leakage alert triggered.

end process

end module

- Java Code

```
package code;
import java.util.*;
class PressureMonitor {
    private double maxPressure;
    private double minPressure;
    private double currentPressure;
    private List<String> pressureLog;
    public PressureMonitor(double maxPressure, double minPressure, double initialPressure) {
        this.maxPressure = maxPressure;
        this.minPressure = minPressure;
        this.currentPressure = initialPressure;
    }
}
```

```

        this.pressureLog = new ArrayList<>();
    }
    public void updatePressure(double changeAmount) {
        currentPressure += changeAmount;
        String message = String.format("Current Pressure: %.1f bar", currentPressure);
        if (currentPressure > maxPressure) {
            message += " | WARNING: High Pressure Detected!";
        } else if (currentPressure < minPressure) {
            message += " | ALERT: Possible Leak Detected!";
        }
        System.out.println(message);
        pressureLog.add(message);
    }
    public void printPressureLog() {
        System.out.println("\nPressure Log:");
        for (String log : pressureLog) {
            System.out.println(log);
        }
    }
}

public class c28_IndustrialPressureSystem {
    public static void main(String[] args) {
        PressureMonitor monitor = new PressureMonitor(10.0, 2.0, 5.0);

        // testcase-VT:
        //     monitor.updatePressure(3.0);
        //     monitor.updatePressure(-4.0);
        //     monitor.updatePressure(-3.0);
        //     monitor.updatePressure(6.0);
        //     monitor.updatePressure(5.0);

        // testcase-FT:
        //     monitor.updatePressure(-2.608417489800224E307);
        //     monitor.updatePressure(1.7836486572462039E308);
        //     monitor.updatePressure(-1.7976931348623157E308);
        //     monitor.updatePressure(3.172399649757028E307);
        //     monitor.updatePressure(-1.450546694021716E308);

        // testcase-VT:
        //     monitor.updatePressure(3);
        //     monitor.updatePressure(4);
        //     monitor.updatePressure(3);
        //     monitor.updatePressure(6);
    }
}

```

```
//      monitor.updatePressure(5);

      // testcase-VT:
      monitor.updatePressure(3.0);
      monitor.updatePressure(4.0);
      monitor.updatePressure(3.0);
      monitor.updatePressure(6.0);
      monitor.updatePressure(5.0);

      monitor.printPressureLog();
  }
}
```

- Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c28_IndustrialPressureSystem.java	100.0 %	6	0	6
> c28_IndustrialPressureSystem		0	0	0
> PressureMonitor	100.0 %	6	0	6

3. Vibration testing instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	106	17,835	17,941
▼ src	0.6 %	106	17,835	17,941
▼ code	0.6 %	106	17,835	17,941
▼ c28_IndustrialPressureSystem.java	97.2 %	106	3	109
> c28_IndustrialPressureSystem	89.3 %	25	3	28
> PressureMonitor	100.0 %	81	0	81

4. Fuzz testing branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	837	842
▼ src	0.6 %	5	837	842
▼ code	0.6 %	5	837	842
▼ c28_IndustrialPressureSystem.java	83.3 %	5	1	6
> PressureMonitor	83.3 %	5	1	6
> c28_IndustrialPressureSystem		0	0	0

5. Fuzz testing instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	106	17,835	17,941
▼ src	0.6 %	106	17,835	17,941
▼ code	0.6 %	106	17,835	17,941
▼ c28_IndustrialPressureSystem.java	97.2 %	106	3	109
> c28_IndustrialPressureSystem	89.3 %	25	3	28
> PressureMonitor	100.0 %	81	0	81

6. Z3 branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	835	840
▼ src	0.6 %	5	835	840
▼ code	0.6 %	5	835	840
▼ c28_IndustrialPressureSystem.java	83.3 %	5	1	6
> PressureMonitor	83.3 %	5	1	6
> c28_IndustrialPressureSystem		0	0	0

7. Z3 instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	103	22,295	22,398
✓ src	0.5 %	103	22,295	22,398
✓ code	0.5 %	103	22,295	22,398
✓ c28_IndustrialPressureSystem.java	94.5 %	103	6	109
> c28_IndustrialPressureSystem	89.3 %	25	3	28
> PressureMonitor	96.3 %	78	3	81

8. Unimproved V-method branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.6 %	5	835	840
✓ src	0.6 %	5	835	840
✓ code	0.6 %	5	835	840
✓ c28_IndustrialPressureSystem.java	83.3 %	5	1	6
> PressureMonitor	83.3 %	5	1	6
> c28_IndustrialPressureSystem		0	0	0

9. Unimproved V-method instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	103	22,295	22,398
✓ src	0.5 %	103	22,295	22,398
✓ code	0.5 %	103	22,295	22,398
✓ c28_IndustrialPressureSystem.java	94.5 %	103	6	109
> c28_IndustrialPressureSystem	89.3 %	25	3	28
> PressureMonitor	96.3 %	78	3	81